PHYS 4027 / 5027 · FALL 2026 · LECTURE 22

Monte Carlo integration and importance sampling

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Last time

- A random number generator is a deterministic recurrence on a state. The seed fixes the whole sequence, which is what makes a run reproducible.
- We turned uniform numbers into samples from other distributions with the inverse transform $x = F^{-1}(u)$, with the Box-Muller transformation for the Gaussian, and with rejection sampling under an envelope.

Today we use random points to compute integrals. We will see why the error falls as $1/\sqrt{N}$ in any number of dimensions, and how to place the points where the integrand is large.

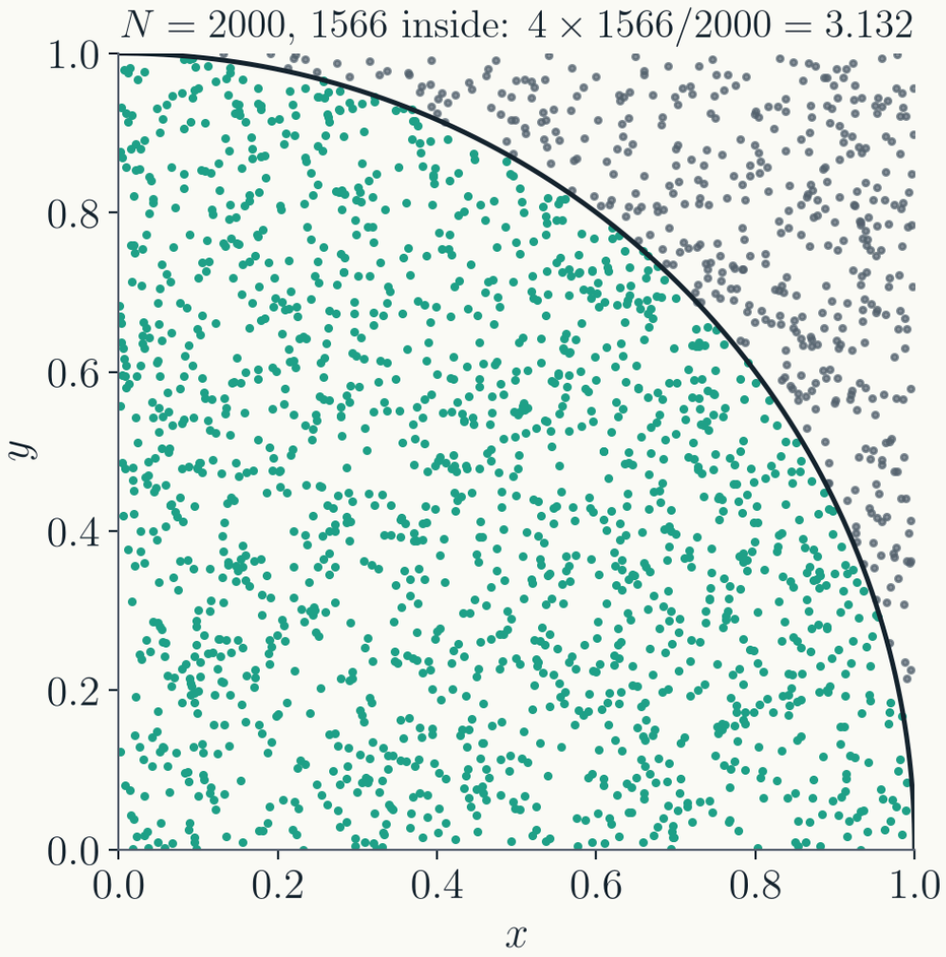
Part I · Monte Carlo integration

Averages over random points

Darts

Throw N points uniformly into the unit square. The fraction that lands inside the quarter circle estimates its area, $\pi/4$, so

$$\pi \approx 4 \times \frac{\text{number inside}}{N}.$$



An integral is an average

$$I = \int_V f \, dV = V \, \langle f \rangle, \qquad \langle f \rangle \approx \frac{1}{N} \sum_{i=1}^{N} f(\mathbf{x}_i),$$

where the points $\mathbf{x}_i$ are drawn uniformly from V.

- The darts are the special case in which f is 1 inside the circle and 0 outside, with $V = 1$.
- The same formula works for any integrand, any region, and any number of dimensions.

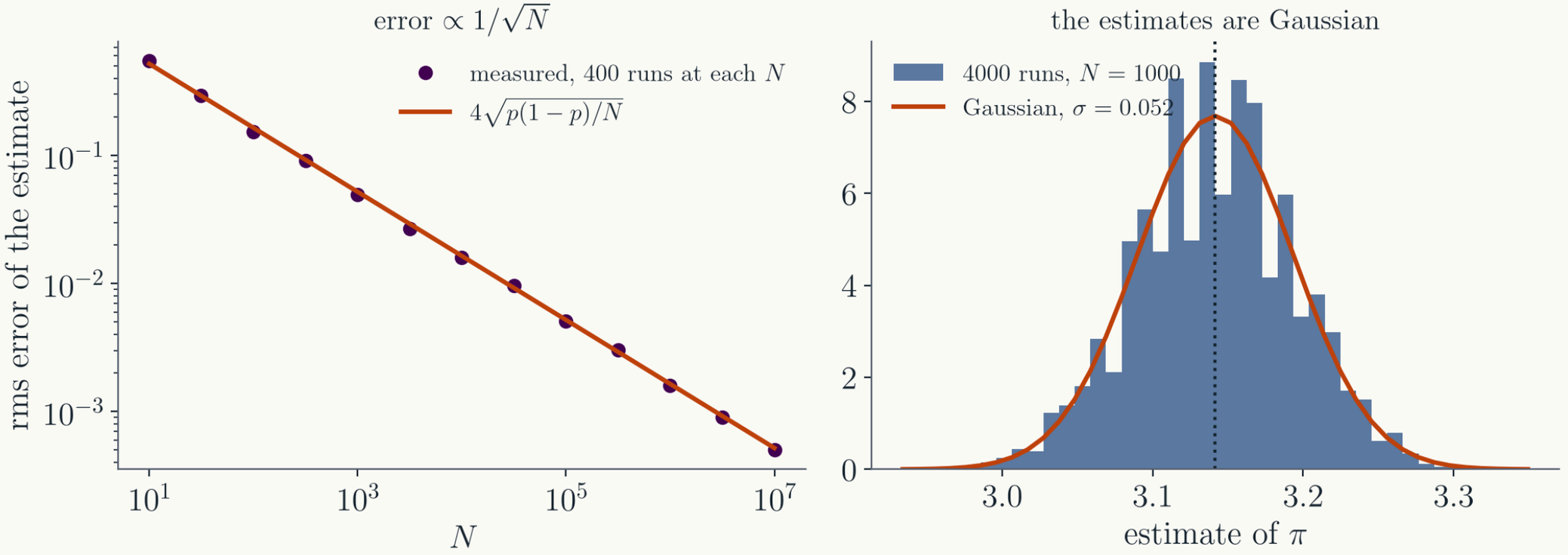
How wrong is it?

Each $f(\mathbf{x}_i)$ is a random number with mean $\langle f \rangle$ and variance $\sigma_f^2 = \langle f^2 \rangle - \langle f \rangle^2$. The average of N independent ones has variance σ_f^2 / N, so

$$I_N = I \pm V \frac{\sigma_f}{\sqrt{N}}.$$

By the central limit theorem the estimates are distributed as a Gaussian around I, so the $\pm$ above is one standard deviation.

Check with the darts



At $N = 1000$ the formula predicts an error of $1.64/\sqrt{1000} = 0.052$. The standard deviation of 4000 independent runs is 0.052.

The error from the same points

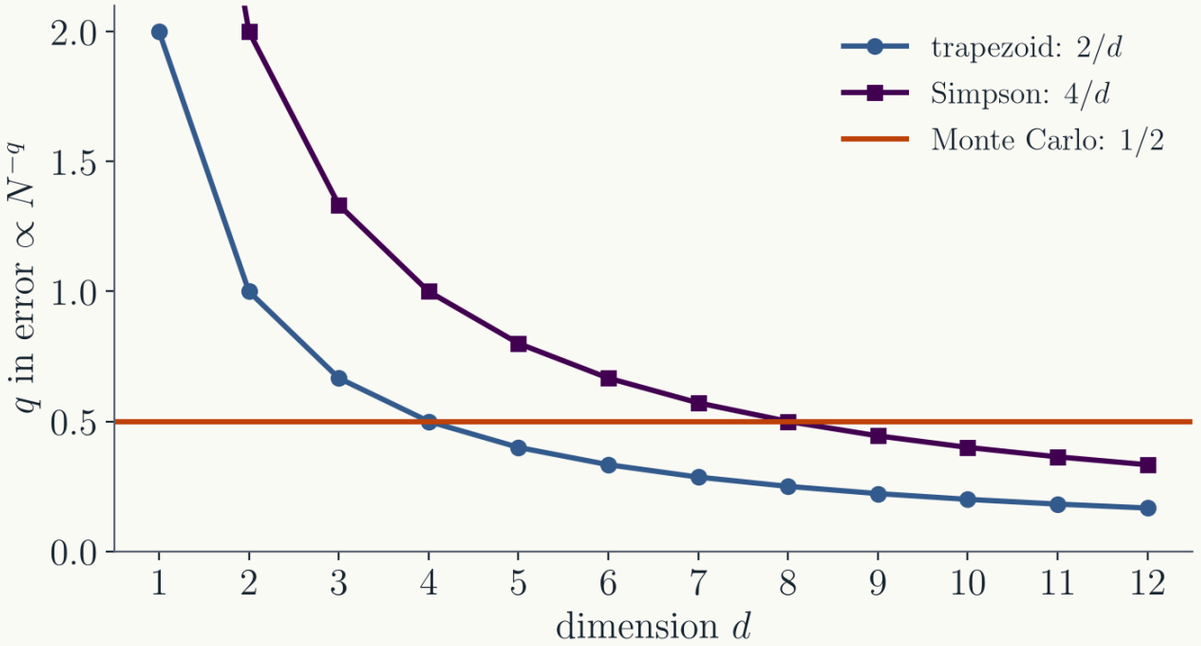
We can estimate σ_f from the same points. If the loop accumulates $\sum f_i$ and $\sum f_i^2$ along the way, then

$$\sigma_f^2 \approx \frac{1}{N}\sum_i f_i^2 - \left(\frac{1}{N}\sum_i f_i\right)^2.$$

- Every run therefore comes with its own error estimate, and a Monte Carlo result should always be quoted with it.
- Because the error falls as $1/\sqrt{N}$, improving the accuracy by a factor of ten requires a hundred times as many points.

Against quadrature

In Lecture 5 we found that the error of the trapezoidal rule scales as h^2 and that of Simpson's rule as h^4. A grid in d dimensions with n points per axis has $N = n^d$ points in total, so $h \propto N^{-1/d}$ and the errors become $N^{-2/d}$ and $N^{-4/d}$. The Monte Carlo error scales as $N^{-1/2}$ in any number of dimensions.

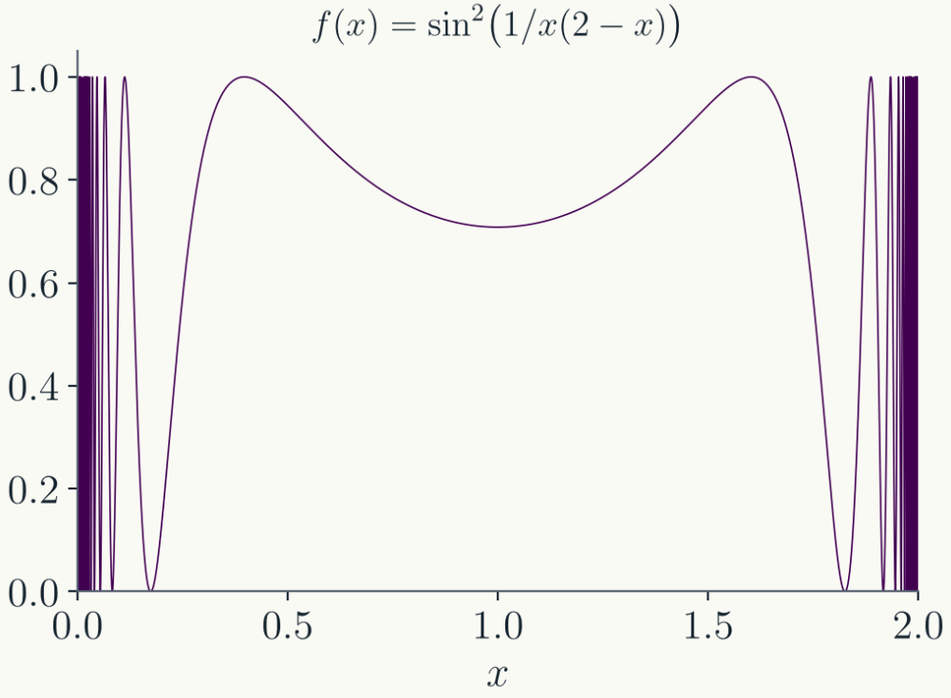


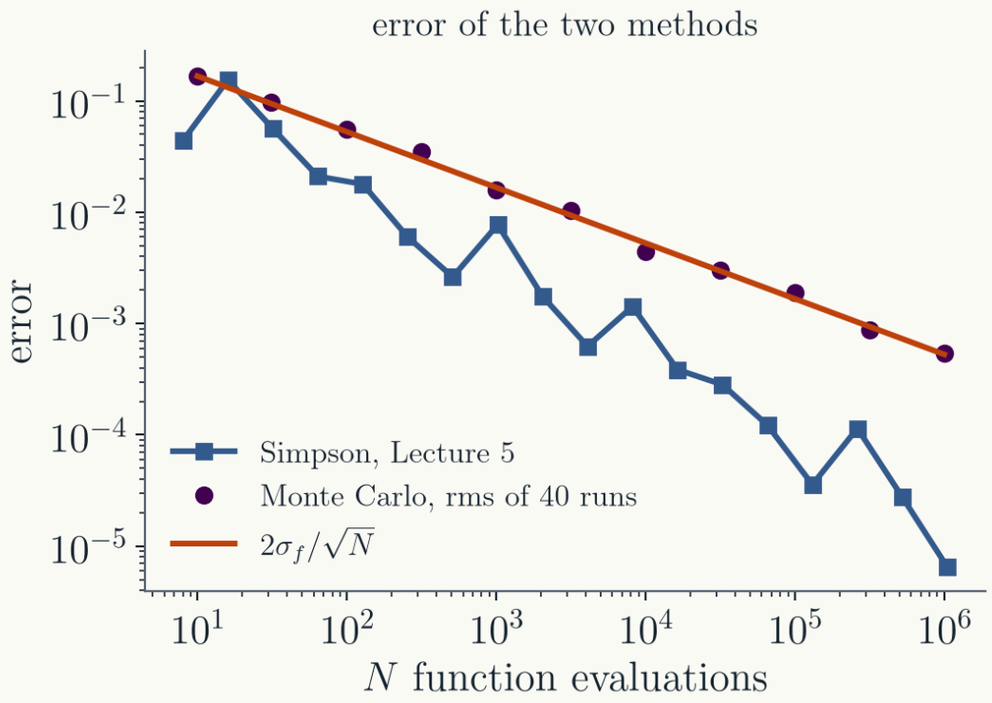
Where the dimensions come from

- The configuration integral of a classical gas of N atoms runs over $3N$ coordinates.
- The partition function of a hundred spins is a sum over 2^{100} states, which is the same problem in discrete form.
- The phase space of a decay into four particles has 8 dimensions, and a path integral has one dimension per time slice.

Every thermal average in statistical mechanics is an integral of this kind, and we will come back to them in the next lectures.

Smoothness is not required





The error formula for Simpson's rule contains f''', and this integrand has no derivatives at the two ends. The Monte Carlo error depends only on $\sigma_f = 0.26$, and it follows the formula.

Finite variance is required

$$\int_0^1 \frac{dx}{\sqrt{x}} = 2, \qquad \langle f^2 \rangle = \int_0^1 \frac{dx}{x} = \infty.$$

- The average still converges to 2, but the error estimate $\sigma_f / \sqrt{N}$ is meaningless, because a single point at $x = 10^{-6}$ adds $1000/N$ to the average.
- An integrable singularity is therefore not good enough for plain Monte Carlo. Part II shows how to remove it.

Part II · Importance sampling

Putting the points where the integrand is

Sampling from a different density

$$I = \int f\,dx = \int \frac{f}{p}\,p\,dx = \left\langle \frac{f}{p} \right\rangle_p .$$

- If we draw the x_i from a density p, using the methods of Lecture 21, and average f/p instead of f, we obtain the same integral.
- The error is now $\sigma(f/p)/\sqrt{N}$. If p has roughly the shape of f, then f/p is nearly constant and its variance is small.
- Uniform sampling is the special case $p = 1/V$.

The recipe

1. Choose a weight $w(x) \geq 0$ that reproduces the peak or the singularity of f and that you know how to sample.
2. Normalize it to a density, $p = w / \int w \, dx$.
3. Draw $x_1, \ldots, x_N$ from p by the inverse transform.
4. Estimate $I \approx \dfrac{1}{N} \displaystyle\sum_i \frac{f(x_i)}{p(x_i)}$, and compute the error from $\sigma(f/p)$ as before.

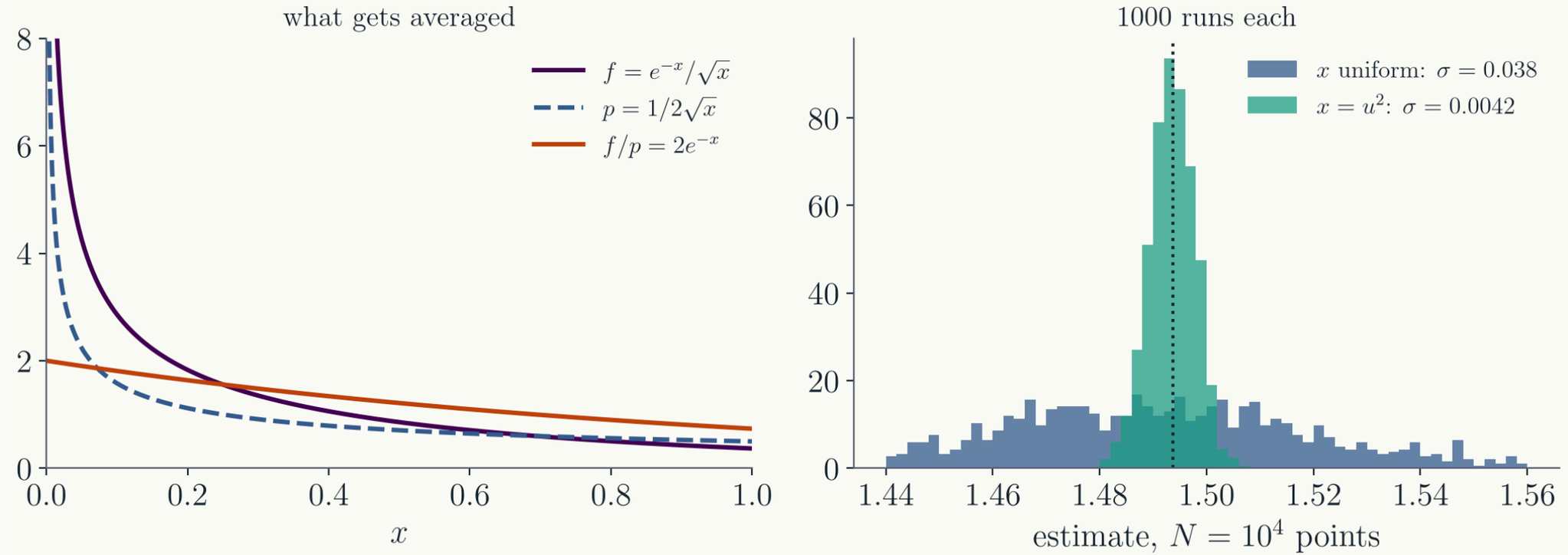
Example: the singularity

$$I = \int_0^1 \frac{e^{-x}}{\sqrt{x}}\,dx = \sqrt{\pi}\,\mathrm{erf}(1) = 1.4936.$$

- With uniform x the variance $\langle f^2 \rangle = \int_0^1 e^{-2x}/x\,dx$ is infinite, so the error estimate cannot be trusted.
- Choose instead $w = 1/\sqrt{x}$. Then $p = 1/2\sqrt{x}$, $F = \sqrt{x}$, and the inverse transform gives $x = u^2$. The ratio $f/p = 2e^{-x}$ lies between 0.74 and 2.

$$I \approx \frac{2}{N}\sum_{i=1}^{N} e^{-u_i^2}.$$

The singularity, side by side



With $N = 10^4$ points and 1000 runs of each method, the spread of the estimates falls from 0.038 to 0.0042. The accuracy improves ninefold, which would otherwise require eighty times as many points.

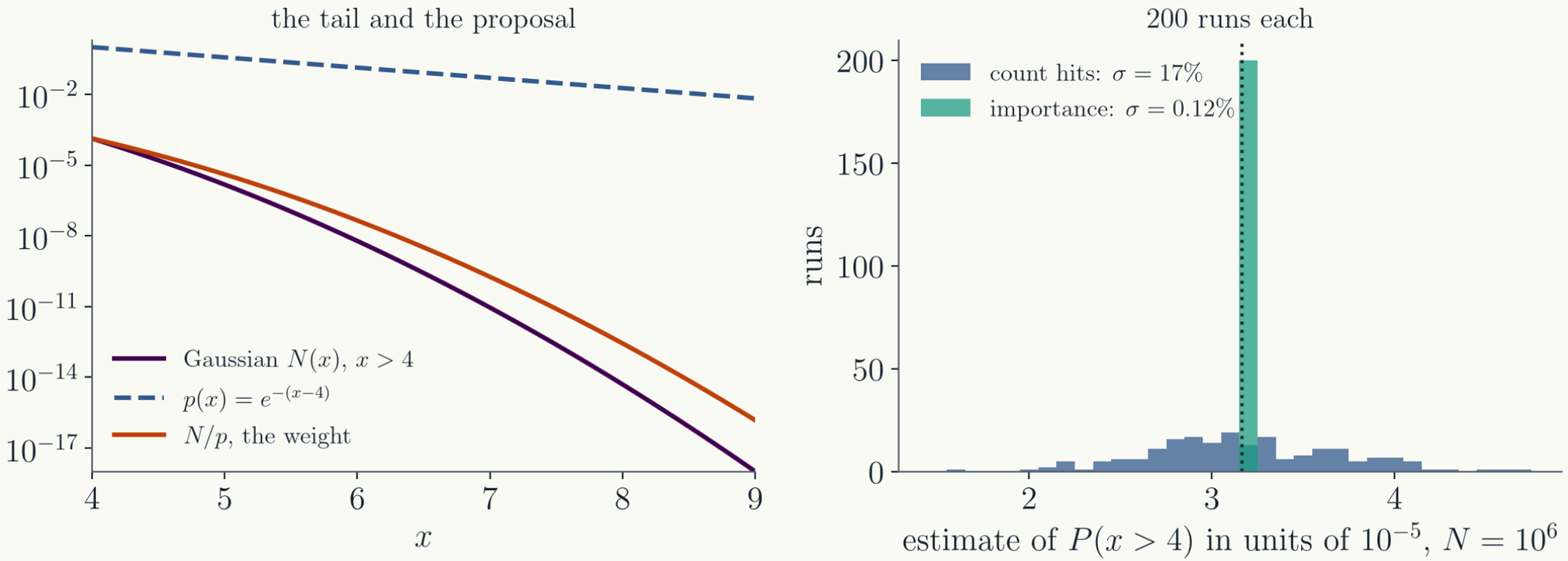
Example: a rare event

The probability that a unit Gaussian variable exceeds 4 is $P = 3.17 \times 10^{-5}$, about one draw in $32{,}000$.

- If we draw $N = 10^6$ Gaussian numbers and count those above 4, we expect 32 ± 6 hits, an error of 17%.
- With importance sampling we draw $x = 4 - \ln(1 - u)$, which is exponentially distributed beyond 4 with density $p = e^{-(x-4)}$, and average

$$\frac{N(x)}{p(x)} = \frac{e^{-x^2/2 + x - 4}}{\sqrt{2\pi}}.$$

The rare event, side by side



With $N = 10^6$ and 200 runs of each method, the relative spread is 17% for counting and 0.12% for importance sampling. Reaching the same accuracy by counting would take $20{,}000$ times as many draws.

Choosing p

- p should have the shape of $|f|$, with the same peaks and the same singularities.
- p must not have lighter tails than f. Wherever p falls off faster than f, the weight f/p grows without bound, the variance is infinite again, and the estimate looks converged while it is wrong.
- p must be cheap to sample, by an inverse transform or a transformation from Lecture 21.

The Boltzmann weight as importance sampling

$$\langle A \rangle = \frac{\int A(\mathbf{x})\, e^{-E(\mathbf{x})/kT}\, d\mathbf{x}}{\int e^{-E(\mathbf{x})/kT}\, d\mathbf{x}}$$

- With uniformly chosen configurations, almost every point has $e^{-E/kT} \approx 0$ and contributes nothing.
- If instead we draw configurations with $p \propto e^{-E/kT}$, the weights cancel and $\langle A \rangle \approx \frac{1}{N} \sum_i A(\mathbf{x}_i)$ is a plain average over the sampled configurations.
- But this p lives in 10^{23} dimensions and its normalization is unknown, so the inverse transform cannot be used. We need a different way to draw from p.

Summary

- A Monte Carlo integral is $I = V\langle f \rangle$ over uniform points, with error $V\sigma_f/\sqrt{N}$ in any number of dimensions. The same points give σ_f, so every result comes with an error bar.
- For smooth integrands quadrature is more accurate below about $d = 4$ to 8 dimensions, and Monte Carlo above that, or whenever f is rough. Monte Carlo requires $\langle f^2 \rangle$ to be finite.
- Importance sampling draws the points from a density p and averages f/p, with error $\sigma(f/p)/\sqrt{N}$. A good p has the shape of f and tails at least as heavy.

Next time: random walks and Brownian motion.